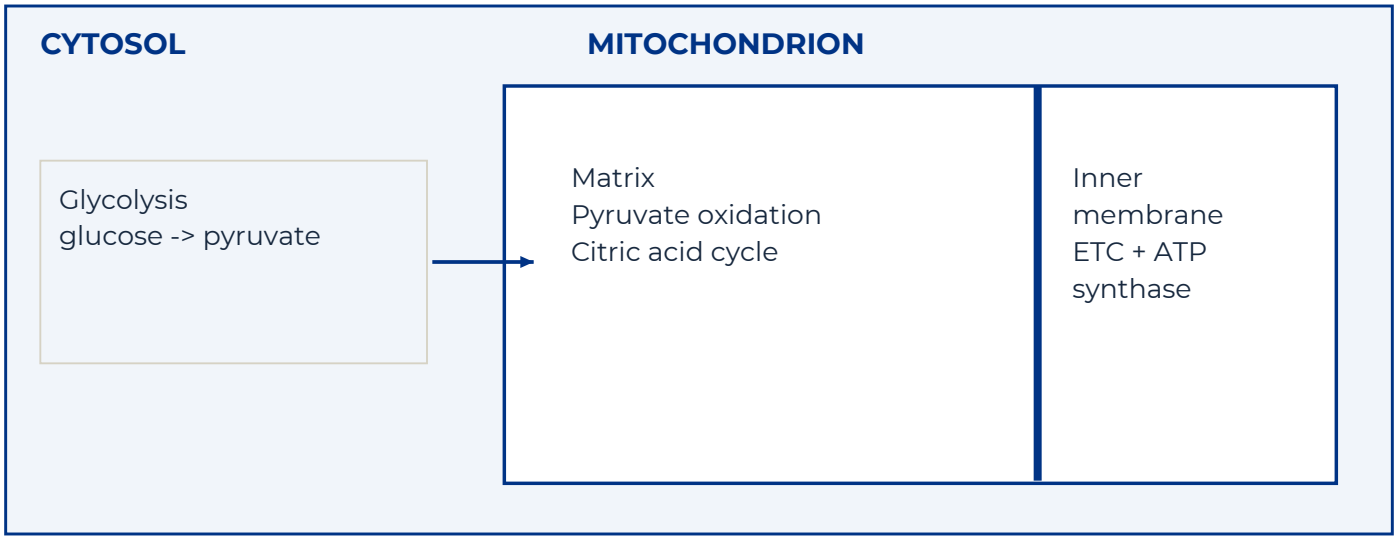
Cellular respiration map

Eukaryotic respiration: keep carbon flow, electron transfer and ATP formation separate.



Stage	Carbon / electrons	ATP mechanism
Glycolysis	Glucose carbon becomes pyruvate; NADH is formed.	Substrate-level phosphorylation: net 2 ATP/glucose.
Pyruvate oxidation + cycle	CO2 is released; NADH and FADH2 carry electrons.	Cycle includes substrate-level phosphorylation.
ETC + ATP synthase	Electron transfer to O2 makes water and drives proton pumping.	Proton return through ATP synthase drives oxidative phosphorylation.

Oxygen and fermentation

O2 is the terminal electron acceptor in aerobic respiration. Fermentation regenerates NAD+ so glycolysis can continue; it does not add ETC-derived ATP.

Avoid a fixed total ATP promise: shuttles, coupling and cell conditions affect aerobic yield. This location map is eukaryotic; prokaryotic locations differ.

Cellular respiration map

Self-check questions and explanations. Refer to the visual guide on the front.

Where do eukaryotic respiration stages occur?

Glycolysis: cytosol. Pyruvate oxidation and citric acid cycle: mitochondrial matrix. Electron transport and ATP synthase: inner mitochondrial membrane, coupled through a proton electrochemical gradient.

Which stages release glucose carbon as CO2?

Pyruvate oxidation and the citric acid cycle release CO2 in the standard aerobic pathway. Glycolysis and the respiratory chain do not directly release glucose carbon as CO2.

What accepts electrons at the end of aerobic respiration?

O2 is the final electron acceptor; it is reduced to water. Oxygen is not directly converted into CO2 in this electron-transfer step.

How do NADH electrons help produce ATP?

Oxidation of fuel reduces NAD+ to NADH. NADH transfers electrons into the chain; favorable electron transfer powers proton pumping. Proton return through ATP synthase drives ATP formation from ADP and phosphate.

What happens after an electron-transport-chain block?

Blocking the chain impairs NADH oxidation and proton pumping; the gradient dissipates and oxidative ATP production falls. Substrate-level ATP formation is a distinct mechanism, though continued metabolism may be constrained.

How does fermentation let glycolysis continue?

Fermentation regenerates NAD+ so glycolysis can continue, giving net 2 ATP per glucose from glycolysis. Fermentation itself does not add respiratory-chain ATP and does not require a mitochondrial ETC.